warp it. And just as the bending energy is stored in the bent bow (until the string is released and feeds the bow's energy into the arrow), so the warping energy is stored in the black hole's warped space. And for a black hole, that energy of warping is so great that it generates the warping.

Warping begets warping in a nonlinear, self-bootstrapping manner. This is a fundamental feature of Einstein's relativistic laws, and so different from everyday experience. It's somewhat like a hypothetical science-fiction character who goes backward in time and gives birth to herself.

This warping-begets-warping scenario does not happen in our solar system hardly at all. Throughout our solar system the space warps are so weak that their energy is minuscule, far too small to produce much bootstrapped warping. Almost all the space warping in our solar system is produced directly by matter—the Sun's matter, the Earth's matter, the matter of the other planets—by contrast with a black hole where the warping is fully responsible for the warping.

Event Horizon and Warped Time

When you first hear mention of a black hole, you probably think of its trapping power as depicted in Figure 5.3, not its warped space.